CODE:

```
import math

def print_board(board):
    for row in [board[i * 3:(i + 1) * 3] for i in range(3)]:
        print("| " + " | ".join(row) + " |")
    print()

def check_winner(board):
    win_combinations = [
        [0, 1, 2],
        [3, 4, 5],
        [6, 7, 8],
        [0, 3, 6],
        [1, 4, 7],
        [2, 5, 8],
        [0, 4, 8],
        [2, 4, 6]
    ]

    for combo in win_combinations:
        if board[combo[0]] == board[combo[1]] == board[combo[2]] != " ":
            return board[combo[0]]

    if " " not in board:
        return "Draw"

    return None

def minimax(board, depth, is_maximizing):
```

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winner = check_winner(board)

if winner == "X":
    return -1
elif winner == "O":
    return 1
elif winner == "Draw":
    return 0

if is_maximizing:
    best_score = -math.inf

    for i in range(9):
        if board[i] == " ":
            board[i] = "O"
            score = minimax(board, depth + 1, False)
            board[i] = " "
            best_score = max(score, best_score)

    return best_score

else:
    best_score = math.inf

    for i in range(9):
        if board[i] == " ":
            board[i] = "X"
            score = minimax(board, depth + 1, True)
            board[i] = " "
            best_score = min(score, best_score)
```

```
        return best_score

def best_move(board):
    best_score = -math.inf
    move = None

    for i in range(9):
        if board[i] == " ":
            board[i] = "O"
            score = minimax(board, 0, False)
            board[i] = " "

            if score > best_score:
                best_score = score
                move = i

    return move

def play_game():
    board = [" "] * 9

    print("You are X. Computer is O.")
    print_board(board)

    while True:
        try:
            move = int(input("Enter your move (0-8): "))
        except ValueError:
            print("Please enter a number.")
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            continue

        if move < 0 or move > 8:
            print("Invalid position.")
            continue

        if board[move] != " ":
            print("Position already occupied.")
            continue

        board[move] = "X"
        print_board(board)

        winner = check_winner(board)

        if winner is not None:
            break

        ai_move = best_move(board)
        board[ai_move] = "O"

        print("Computer places O at position", ai_move)
        print_board(board)

        winner = check_winner(board)

        if winner is not None:
            break

    if winner == "Draw":
```

```
        print("It's a Draw!")
    else:
        print("Winner is:", winner)
```

play_game()



